Mermaid.js

Diagrams as Code, Right in Your Repository

v10.9+ | mermaid.js.org ▶ Watch Showcase Video

1. The Problem

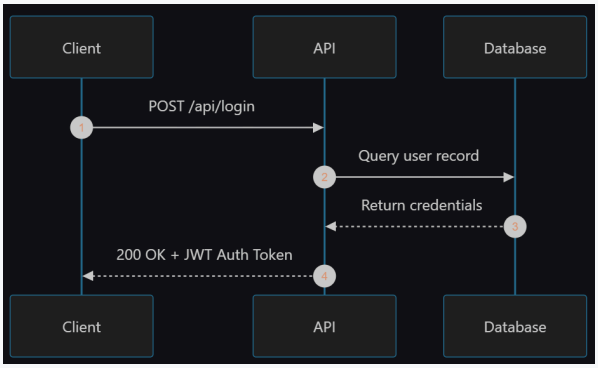
Architecture diagrams and flowcharts rot rapidly because they are typically created in external GUI tools (Visio, Miro, Lucidchart). Because these disconnected, proprietary files cannot be reviewed in pull requests, they immediately fall out of sync when the underlying code ships.

2. Zero-Download Setup & First Result

You do not need to download or install anything. GitHub and GitLab natively compile Mermaid blocks into visuals.

- **Default VS Code (Built-in):** VS Code's default Markdown Preview renders diagrams in real-time side-by-side as you type.
- **Mermaid Extension (Optional):** The *"Mermaid Preview" (or "Mermaid", the version that requires an account)* extension simply adds quality-of-life developer features: syntax highlighting, auto-complete, and live linting (error squiggles) inside the code block.

````
```mermaid
sequenceDiagram
  autonumber
  Client->>API: POST /login
  API->>DB: Query user
  DB-->>API: Return hash
  API-->>Client: 200 OK + JWT
```
````



3. Why This Tool? (vs. Alternatives)

- **Vs. GUI Tools (Lucidchart/Draw.io):** Mermaid offers plaintext version control. Changes to arrows and boxes show up as readable git diffs that engineers can review in PRs alongside code.
- **Vs. AI Assistants alone:** LLMs can write code, but Mermaid provides the standardized visual engine that automatically renders that code inside your developer portal.
- **Vs. PlantUML:** PlantUML requires a local Java runtime and Graphviz installation. Mermaid runs purely on standard web technologies with zero backend overhead.

4. When to Use It

✅ Where it Excels

System flowcharts, API sequences, and ERDs that must evolve with the codebase. Best for teams wanting documentation to undergo code review.

⚠️ Where it Falls Short

Not suitable for pixel-perfect marketing slides requiring precise drag-and-drop aesthetics, or massive network topologies (100+ nodes).

Cost & Limits: 100% Free for Developers

The entire local workflow (VS Code rendering, unlimited diagram complexity, and GitHub integration) is entirely open-source and free. Paid tiers only exist for their separate cloud product ("Mermaid Chart") for enterprise collaboration, which developers do not need for this workflow.